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Numerical Simulation of Lubrication Film in Piston Ring of Fuel Dispenser Flowmeter Based on Hydrodynamic Lubrication

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***Abstract*—Hydrodynamic lubrication is an ideal state of lubrication in that friction and wear hardly occur. The friction pair of piston ring-cylinder liner is one of the most important friction pairs in a flowmeter. With the development of flowmeter aiming at high-precision and high-stability, how to make the piston ring-cylinder liner system maintain a good lubrication condition has great importance in terms of reducing the friction and wear. In this paper, a hydrodynamic lubrication mathematical model on the surface of piston ring was established. And the finite difference method(FDM) and successive over relaxation method(SOR) were adopted to solve the mathematical model numerically. The dimensionless average pressure of the lubrication film was taken as the criterion to estimate the hydrodynamic lubrication performance of the piston ring, and the effects of different flow rate conditions, different fluid viscosities and different minimum film thicknesses on the lubrication film pressure distribution were investigated respectively. The numerical analysis results reveal that: (a) the average pressure decreases with the increase of minimum film thickness, under the condition of the same minimum film thickness, when the medium is diesel, the average pressure is greater than the average pressure when the medium is gasoline; (b) the lubrication film pressure increases with the increase of flow rate, and the flow regime of the lubrication film changes with the change of flow rate; (c) as the viscosity increases, the pressure increases as well, for the same leakage clearance, the higher the viscosity, the larger load-carrying capacity of the clearance lubrication film. These results provide a guidance for lubrication design of piston ring-cylinder liner friction pair.**

Keywords-lubrication film; piston ring; flowmeter; finite difference method; hydrodynamic lubrication (key words)

I. INTRODUCTION

The dispenser is a fuel-pumping and measurement device used in the service station. Flowmeter is the core component of the dispenser metering system that measures the fuel volume which is to be given to the customers. At present, most fuel dispenser flowmeters are positive-displacement piston types. The pistons are activated by the flow of fuel from the pumping system and are mounted in parallel cylindrical liners. A case of piston ring-cylinder liner in a flow meter system is shown in Figure 1. In this case, the piston rings are pushed by the fuel and reciprocate in the cylinder liner. There is a small sealing clearance between the piston ring and the cylinder liner. The clearance is necessary to ensure flexible movement of moving parts to reduce pressure loss and wear, but also to make the internal leakage as little as possible. In theory, the clearance design value should be zero, but in fact it is difficult to achieve. For example, the clearances between rotor and shell of waist wheel flowmeter and oval gear flowmeter are generally designed in 0.1mm or so, generally not more than 0.15mm[1]. A simple model of a piston ring-cylinder liner friction pair is illustrated in Figure 2[2].

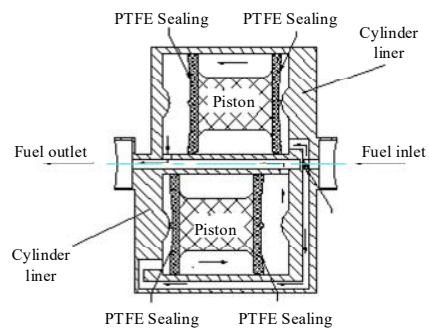

Figure 1. Piston ring-cylinder liner in a flow meter system

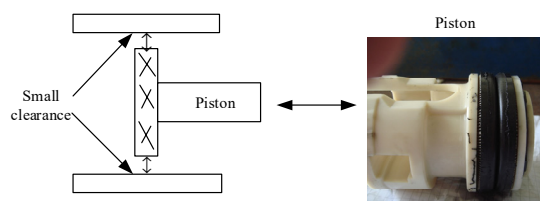

Figure 2. A simple model of a piston ring-cylinder liner friction pair

In general, the internal leakage directly affects the measurement accuracy[3]. The internal leakage is related to many factors, such as the principle of measurement, manufacturing and assembly, the characteristics of medium parameters, the flow regime and so on. Recently, many attentions have been paid for the mechanism of internal leakage. Weng et al.[4] discussed the effect of viscosity on the performance of turbine flow-meter, and improved the flow-meter model of current turbine flow-meter by the dynamics of viscous fluid. Xu et al.[5] investigated the effects of viscosity and density changing on measurement accuracy based on positive displacement meter. Wang et al.[6] analyzed the effects

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of fluid physical parameters and differential pressures on measuring accuracy, and pointed out that to obtain a precise measuring accuracy value needed to amend the temperature and pressure. Cheng [7] studied the effects of mechanical wear, temperature and pressure on the leakage of positive displacement meter, and deduced a calculation formula of measuring accuracy caused by leakage based on the laminar flow condition. Morton et al. [8] performed an experiment to study the relationship of the leakage and the flow rate, the density, the viscosity, the quality of piston, the piston lubrication hole, the piston ring groove surface coating, the connection length for the positive displacement flow-meter (rotary piston). Baker et al. [9] made a comprehensive review of the positive displacement flow-meter (rotary piston), and deduced a calculation formula of leakage when leakage flow regime was laminar. Wang et al. [10] used finite element method to analyze the flow-meter, and studied the measuring response characteristics when bubble in many sizes or in different positions, provided a reference for multiphase measurement of flow-meter. Zhao et al. [11] deduced the flow calculation formula of different flow regimes by the theory of boundary layer.

In addition, the clearance increases by the wear of piston when the piston meter in long-term application. In the direction of movement, the clearance between the piston ring and the cylinder liner is in the form of large to small. Under this circumstance, the fluid can be continuously pumped into this clearance, creating an oil film pressure, forming hydrodynamic lubrication [12]. And the amount of leakage is the part of the fluid volume which was used to establish the pressure of the oil film. However, from the above research, we can find that there is no study to analyze the lubrication film in piston ring of flowmeter and the formation mechanism and the pressure of the oil film are not clear. But they are of vital importance to optimize the clearance and improve the measurement accuracy. Therefore, this research has developed a hydrodynamic lubrication mathematical model on the surface of piston ring based on the steady-state Reynolds' equation; the effects of different flow rate conditions, different fluid viscosities and different minimum film thicknesses on lubrication film pressure were evaluated by numerical calculation. The analytical model was introduced in section 2. In section 3, we analyzed the results and made a discussion. Summary for the whole was given in section4.

II. ANALYTICAL MODEL

In this section, we first deduce the formula for calculating the velocity of piston ring. Then, we present the formation mechanism of the lubrication film pressure. Finally, we shall establish hydrodynamic lubrication mathematical model on the surface of piston ring based on Reynolds' equation.

A. Analysis of Piston Ring Motion Velocity

During the working process of flowmeter, the piston transfers the reciprocating motion to the crankshaft perpendicular to the direction of the piston movement, and the crankshaft is connected with the common transmission shaft through a pair of gear pairs. The displacement of the transmission shaft per revolution is q, so when output flow is Q, the number of rotations of the transmission shaft can be derived by the following equation:

$$N = \frac{Q}{q} \tag{1}$$

The maximum stroke of the piston is s_0, so the displacement of piston ring for each working cycle can be obtained:

$$S = 2Ns_0 \tag{2}$$

The motion velocity of piston ring could be obtained further:

$$U = \frac{S}{T_0} \tag{3}$$

Where T_0 is the rotation period of the transmission shaft.

B. Formation Mechanism of the Lubrication Film Pressure

The Reynolds' equation is a classical equation that describes the theory of hydrodynamic lubrication. Since it was put forward in 1886, it has been widely used in fluid lubrication calculation. This equation reveals the formation mechanism and distribution law of the lubrication film pressure between two relative moving surfaces, which lays the foundation for the modern fluid lubrication theory[13]. The general form of the Reynolds' equation is as follows:

$$\begin{aligned} &\frac{\partial}{\partial x}\left(\frac{\rho h^3}{\eta}\frac{\partial p}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\rho h^3}{\eta}\frac{\partial p}{\partial y}\right) \\ &=6\left(U\frac{\partial \rho h}{\partial x}+V\frac{\partial \rho h}{\partial y}+2\frac{\partial \rho h}{\partial t}\right) \end{aligned} \tag{4}$$

where x and y represent coordinates in a global Cartesian coordinate system parallel to the surface of the friction pair; is the density of the fluid; is the coefficient of kinetic viscosity; h is the thickness of the lubricant film; U is the relative velocity in the x axis direction; V is the relative velocity in the y axis direction; and t is the time entry.

The left-hand side of the Reynolds' equation shows the change in the lubricating film pressure along the coordinates on the lubricating surface, the right-hand side represents the causes of pressure generation[14] and the three terms correspond to the following three mechanisms of pressure generation are illustrated in Figure 3.

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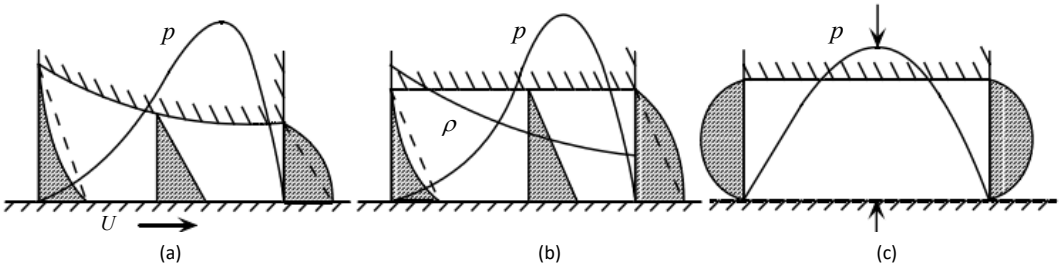

Figure 3. Mechanism of pressure generation: (a) wedge effect;(b) variable density effect;(c) squeeze effect

Specifically, the $U\rho(\partial h/\partial x)$ and $V\rho(\partial h/\partial x)$ are directly obtained from the wedge effect, the $U\rho(\partial h/\partial x)$ and $V\rho(\partial h/\partial x)$ represent the variable density effect, and the $\rho(\partial h/\partial t)$ is the squeeze effect. Generally, the wedge effect and squeeze effect are the main factors that generate lubricating film pressure.

C. *Hydrodynamic Lubrication Mathematical Model on the Surface of Piston Ring*

Among these three effects, the wedge effect is most commonly seen, for example, in journal bearings[15]. The squeeze effect plays an important role in the small-end bearings of crank rods. In this paper, we consider the lubricant is Newtonian fluid, incompressible, and with constant viscosity and density. These assumptions are normal ways similar to that in References[16-19]. Therefore, the squeeze effect and variable density effect can be ignored in this research. The lubrication film pressure in a piston ring-cylinder liner friction pair in the stationary state is given by the following equation, which is derived from Equation (4).

$$\frac{\partial}{\partial x}\left(\frac{h^3\partial p}{\eta\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{h^3\partial p}{\eta\partial y}\right)=6U\frac{\partial h}{\partial x} \tag{5}$$

where the x axis is in the axial direction and y axis is taken in the circumferential direction of the piston; $p(x, y)$ is the oil film pressure; $h(x, y)$ is the oil film thickness; U is relative velocity of the friction pair; and η is coefficient of dynamic viscosity.

The contacting model of two planes with relative sliding velocity is displayed in Figure 4a. The upper plane is the inner surface of the fixed cylinder, and the lower plane is surface of piston ring and moving with a relative velocity of U to the upper plane. The pressure generation due to the fluid being driven from the thick end to the thin end of the wedge-shaped fluid film by the surface movement. Figure 4b shows that piston ring is expanded in the circumferential direction. Because the circumferential length of the piston ring is much larger than the width of the piston ring, in order to facilitate the calculation, we can take a portion of the circumferential length, and then set: $L/B=1.5$. Where the B represents the width of the piston ring and the L is a portion of circumferential length of the piston ring.

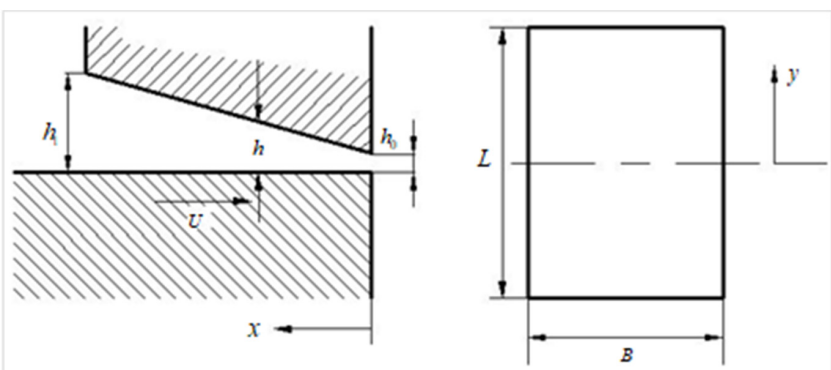

Figure 4. The wedge-shaped model: (a) contacting model of two planes with relative sliding velocity;(b) piston ring expanded in circumferential direction

From the Figure 4, the equation of oil film thickness can be derived. It can be expressed in the following form:

$$h=h_0\left(1+K\frac{x}{B}\right) \tag{6}$$

where $K=(h_1-h_0)/h_0$; h_1 is the maximum film thickness; h_0 is the minimum film thickness. As explained in Reference[20], $h_1/h_0=2.2$.

For the sake of convenience, the above Equation(5) need to be nondimensionalized firstly, the following dimensionless coordinates X and Y, dimensionless film thickness H and

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dimensionless pressure P are defined as followed. The other symbols are the same as before:

$$X=\frac{x}{B},Y=\frac{y}{L},H=\frac{h}{h_0},P=\frac{p}{p_0} \tag{7}$$

where p_0 is ambient air pressure, usually the value is 0.1013MPa; Then, the Equation(5) and Equation(6) can be dimensionless to Equation(8) and Equation(9), respectively.

$$\frac{\partial^2 P}{\partial X^2}+\alpha\frac{\partial^2 P}{\partial Y^2}=\lambda\frac{\partial H}{\partial X} \tag{8}$$

Where $\alpha=B^2/L^2$; $\lambda=6U\eta B/H^3h_0^2p_0$.

$$H=1+K\frac{X}{B} \tag{9}$$

Boundary conditions are required to solve Equation(4). In this paper, we adopt the Reynolds' boundary which the lubrication film is assumed to terminate at a position in the divergence region at which pressure gradient is zero, simultaneously, is given by Figure 5. The pressures at the boundaries of inlet (x = -B) and outlet (x = x') are shown in Equation(10). And the pressures at other two boundaries of (y = -$L/2$ and y = $L/2$) are equal and change periodic from row to row, as illustrated in Equation(11).

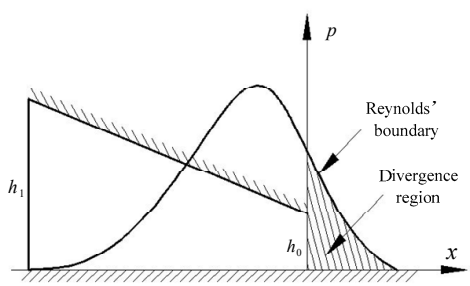

Figure 5. Boundary conditions for the lubrication film

$$p(x=-B,y)=p_0,p(x=x',y)=p_a,\frac{\partial p}{\partial x}(x=x',y)=0 \tag{10}$$

$$\begin{aligned}&p(x,y=-L/2)=p(x,y=L/2)=0,\\&\frac{\partial p}{\partial y}(x,y=-L/2)=\frac{\partial p}{\partial y}(x,y=L/2)=0\end{aligned} \tag{11}$$

The Equation(8) is a two-dimensional equation, however, it cannot be solved analytically in general, but must be solved numerically. The finite difference method [21] is adopted in this research. In order to obtain the stable and accurate numerical solutions, the number of circumferential grids and axial grids are set to 160 and 100, respectively. The discretization of the Equation(8) leads to a set of linear algebraic equations, by solving these simultaneously, the pressures at respective nodal points are obtained. The successive over relaxation method(SOR) is used to solve these simultaneous equations, thereby, pressure at all the nodal points can thereby be found in a finite number of operations. The over-relaxation coefficient of $\varepsilon=1.75$ is chosen and the converging condition of this iterative method is confirmed by Equation(14).

$$\frac{\sum_{i,j}\left|p(i,j)_k-p(i,j)_{k-1}\right|}{\sum_{i,j}\left|p(i,j)_k\right|}\le 10^{-6} \tag{12}$$

where $p(i,j)_k$ are the results in the kth calculation, $p(i,j)_{k-1}$ are those obtained in the previous calculation, and 10^{-6} is a sufficiently small allowable error.

III. Results and Discussion

In this section, we define the relevant parameters of the above equations firstly. Then, the dimensionless average value P_{av} of pressure distribution is chosen as an index to evaluate the load-carrying capacity of lubrication film, and the effects of different gasoline flow rate conditions on the lubrication film pressure distribution were investigated when the minimum film thickness is 50 μm. In addition, the influences of different fluid viscosities (gasoline or diesel) on the pressure distribution of lubricating film were studied under the same flow conditions when the minimum film is also 50 μm. At last, we studied the effects of different minimum film thicknesses on the average pressure, and the reference for the optimal design of clearance was put forward.

A. Parameter Definition

First of all, values of geometric parameters listed in Table I will be applied to this study.

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TABLE I. VALUES OF GEOMETRIC PARAMETERS

Geometric parameters	Definitions	Values
B	The width of the piston ring (mm)	12
L	A portion of circumferential length (mm)	18
h_0	The minimum film thickness (μm)	1-100
s_0	The maximum stroke of the piston (mm)	39
q	The displacement of the transmission shaft per revolution (L/r)	0.5

And then, gasoline and diesel are often measured in fuel dispenser flowmeter. The properties of these two medium at room temperature are shown in Table II.

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TABLE II. TEMPERATURE

Medium	Density ρ $(10^3 kg/m^3)$	Dynamic viscosity η $(mPa \cdot s)$
Gasoline	0.738	0.527
Diesel	0.834	4.25

In addition, there is a one-to-one relationship between flow rate, outlet pressure and the motion velocity of piston ring. The measurement range of the flowmeter is 2*L/min*-80*L/min*. According to the verification standard[22], eight flow points are taken for numerical calculation in this paper. As illustrated in Table III.

TABLE III. THE VALUES OF FLOW RATE, OUTLET PRESSURE AND MOTION VELOCITY

Test number	Parameters		
	Flow rate Q (*L/min*)	Density ρ (*MPa*)	Dynamic viscosity η (*m/s*)
1	10	0.1021	0.026
2	20	0.1044	0.052
3	30	0.1082	0.078
4	40	0.1136	0.104
5	50	0.1204	0.130
6	60	0.1288	0.156
7	70	0.1388	0.182
8	80	0.1503	0.208

B. Effects of Different Gasoline Flow Rate Conditions on Average Pressure

The flowchart of piston ring hydrodynamic lubrication calculation is shown in Figure 6. The numerical results of dimensionless average pressure are shown in the following Figure 7.

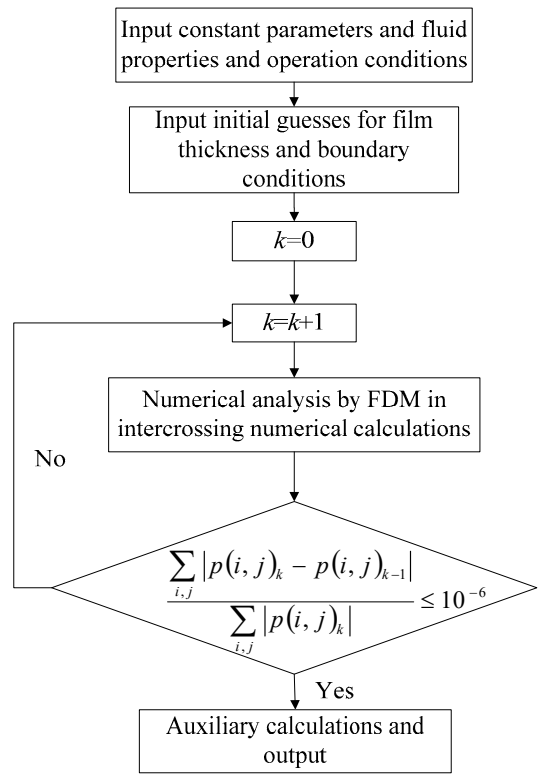

Figure 6. The flowchart of computational procedure

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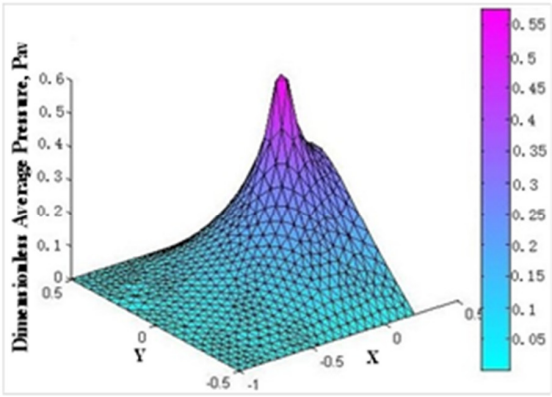
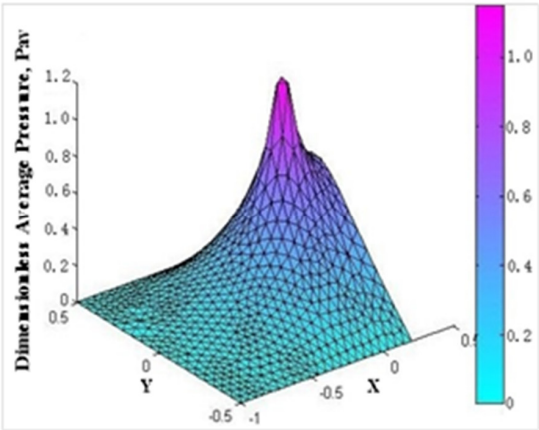
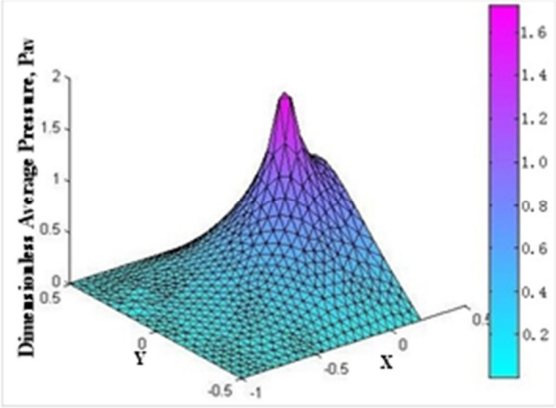
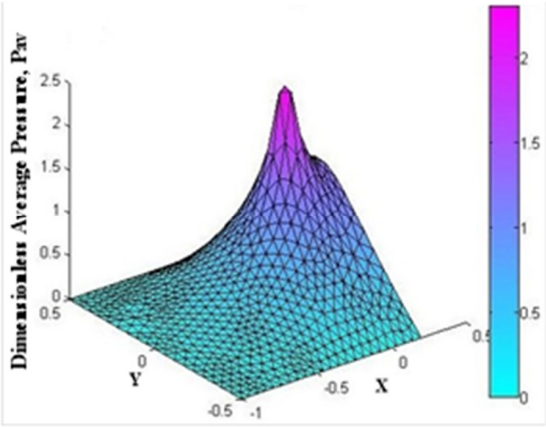


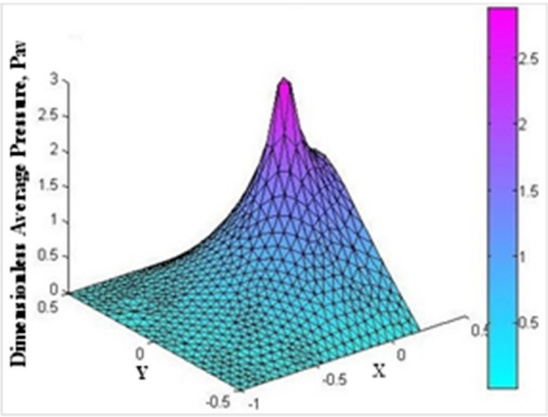


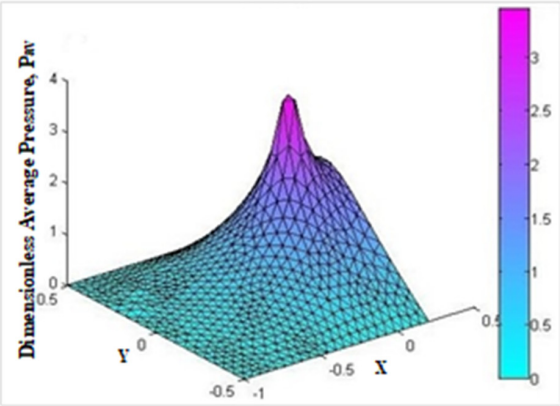
(a) Q=10L/min (b)Q=20L/min





(c) Q=30L/min (d) Q=40L/min





(e) Q=50L/min (f) Q=60L/min

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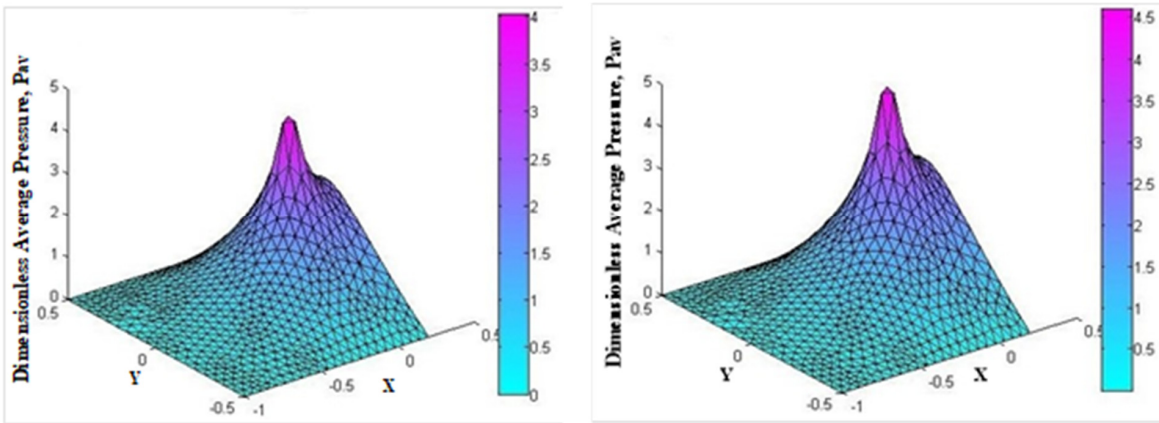

(g) Q=70L/min (h) Q=80L/min

Figure 7. Lubrication film pressure distribution of different gasoline flow rate conditions

Figure 7 shows eight dimensionless hydrodynamic pressure distributions generated by different flow rate conditions. The axial coordinate value of the piston ring takes dimensionless, that is, coordinate value 0 is defined as the axial midpoint of the piston ring, the coordinate value -0.5 means that the lubricant enters the inlet end of the piston ring, coordinate value 0.5 means that the lubricating oil flows out of the outlet end face of the piston ring. The flow direction of the lubricant is parallel to X axis. As illustrated, the maximum dimensionless average pressure appears at the right edge of the piston ring. The reason is that additional hydrodynamic pressure is generated through the convergent wedge in the region. However, at the end of the edge, there is a divergence region due to the rupture of oil film. But the cavitation can occur in this region according to the Reynolds's boundary conditions.

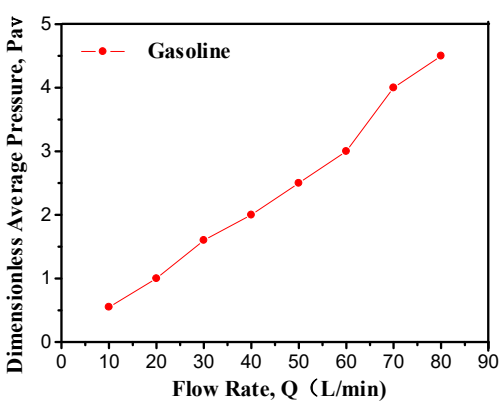

Figure 8. The relationship between dimensionless average pressure and flow rate

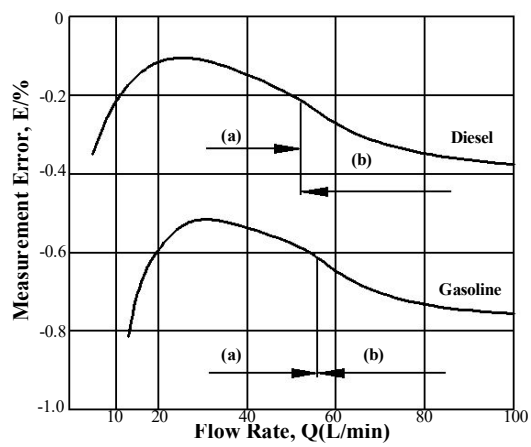

Figure 9. The relationship between measurement error and flow rate: (a)laminar flow; (b)transitive and turbulent flow[23]

It can be seen from Figure 8 that the lubrication film pressure increases with the increase of flow rate. More specifically, the dimensionless average pressure value is less than 1 when the flow rate is less than 20*L/min*, at the same time, it can be seen from Figure 9 that the measurement error is larger, which is due to the insufficient oil film load-carrying capacity, resulting in increased leakage in the clearance. Then, when the flow rate is between 30*L/min* and 55*L/min* the oil film pressure increases and has sufficient load-carrying capacity, Figure 9 shows that in this case the measurement error is small, and remain stable, the flow regime of clearance is laminar flow. In addition, when the flow rate is greater than 60L/min, the oil film pressure increases sharply and exceeds the maximum nominal pressure of the flowmeter, resulting in severe wear to the piston ring seal, therefore, the Figure 9 shows the flow regime of clearance is transitive and turbulent flow and the measurement error increases rapidly under this condition.

C. Effects of Different Fluid Viscosities on the Average Pressure

At present, the domestic and international market of fuel filling industry mainly sells gasoline and diesel, the biggest difference between gasoline and diesel is that the viscosity is

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different. Typically, the viscosity of diesel is 10 times as much as gasoline. In lubrication theory, the most important physical property of a lubricant is its viscosity. Therefore, we mainly study the influences of different fluid viscosities (gasoline or diesel) on the dimensionless average pressure distribution of lubricating film under the same flow conditions. In order to facilitate the study, we selected four flow points (namely 30*L/min,* 40*L/min*, 50*L/min* and 60*L/min*) for comparative analysis. Figure 10 shows the oil film pressure distribution at four different flow rates when the medium of clearance is diesel. It can be seen from Figure 10 that the distribution form of oil film is basically the same under different flow rates, but the value of oil film pressure is different. As shown below, the lubrication film pressure increases with the increase of flow rate and the maximum dimensionless average pressure appears at the right edge of the piston ring.

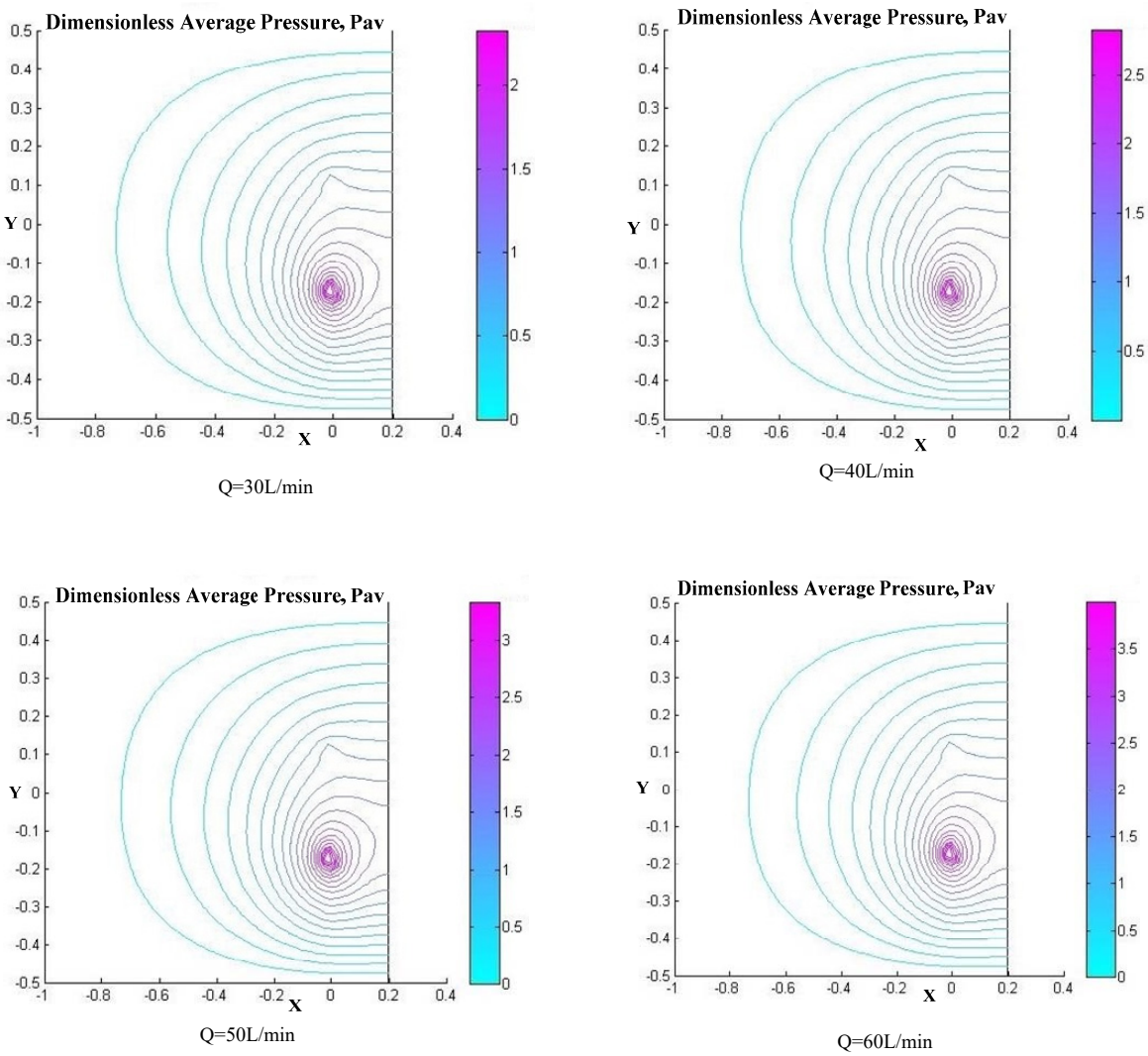

Figure 10. The contour map of Lubrication film pressure distribution

Based on Figure 8 and Figure 10, we can get the following Figure 11. As shown in Figure 11, when the medium is diesel, the lubrication film pressure at each flow point is greater than the lubrication film pressure when the medium is gasoline. The results show that the lubrication film pressure increases with the increase of viscosity. At the same time, it can be seen from Figure 10 that the error curve moves in the positive direction of the Y axis with the increase of fluid viscosity. This is because for the same leakage clearance, the higher the viscosity, the larger the clearance lubrication film load-carrying capacity, leading to the reduction of clearance leakage and measurement error. However, it is necessary to study the specific mathematical relationship between the viscosity and lubrication film load-carrying capacity in next research.

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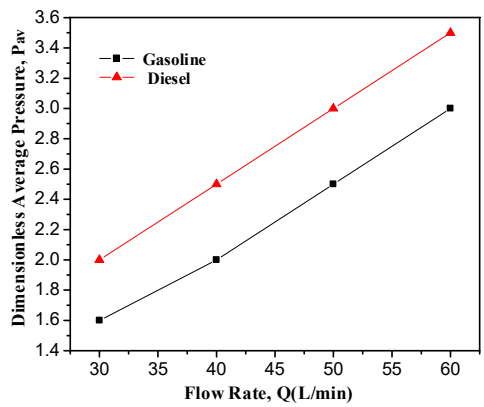

Figure 11. The change of lubrication film pressure under different medium conditions

D. *Effects of Different Minimum Film Thicknesses on the Average Pressure*

In this part, we studied the effects of different minimum thicknesses on the dimensionless average pressure distribution of lubricating film. For the sake of convenience, six different minimum film thicknesses (namely 30 μm , 40 μm , 50 μm , 60 μm , 70 μm and 80 μm) were selected to analyze. In fact, the refueling flow of gasoline is generally not more than 40*L/min* during the working process of the fuel dispenser. We set the flow rate to 40*L/min*, the relationship between average pressure and minimum film thickness was obtained in Figure 12. As shown in Figure 12, whether the medium is diesel or gasoline, the average pressure decreases with the increase of minimum film thickness. Under the condition of the same film thickness, when the medium is diesel, the average pressure is greater than the average pressure when the medium is gasoline. However, as the maximum film thickness increases, the difference between the two gradually decreases.

But more importantly, it can be found from Figure 11 and Figure 12 that whether at the same minimum film thickness or the same flow conditions, the average pressure of the diesel medium is greater than the average pressure of the gasoline medium. This indicates that when filling gasoline, the lubrication film load-carrying capacity of piston ring-cylinder liner clearance is different from that when filling diesel. This provides further evidence that when filling diesel and gasoline, the piston ring-cylinder liner clearance of flowmeter should be set to different parameters, respectively. But in fact, in the current practical engineering applications, regardless of filling diesel or gasoline, the fuel dispenser flowmeters are the same type. Based on the above analysis, this design is unreasonable. In this paper, the clearance parameters were optimized. Specifically, as is clearly seen from the Figure 12, when the flowmeter operates in a stable condition, the dimensionless average pressure of lubrication film in piston ring-cylinder liner clearance is usually 2.5. Under this condition, the maximum film thickness corresponding to the diesel medium is between 45 μm and 47.5 μm , and the maximum film thickness corresponding to the gasoline medium is between 50 μm and 52.5 μm . It is worth noting that the minimum film thickness depends largely on the size of the clearance value. To sum up, we can come to the conclusion that the medium factor should be taken into account when designing the clearance value of the fuel dispenser flowmeter. The clearance of flowmeter used to measure the gasoline is smaller than that of the flowmeter used to measure the diesel. Certainly, the specific clearance values need further theoretical and experimental studies.

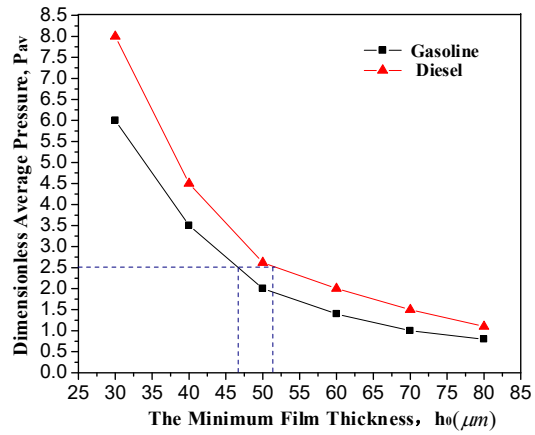

Figure 12. The relationship between dimensionless average pressure and minimum film thickness

IV. CONCLUSIONS

In this paper, the friction pair of piston ring-cylinder liner in a flowmeter was selected as the research object. A hydrodynamic lubrication mathematical model on the surface of piston ring was developed to investigate the effects of different flow rate conditions, different fluid viscosities and different minimum film thicknesses on the lubrication film pressure distribution, respectively. The dimensionless average pressure was calculated and chosen as an index to evaluate the load-carrying capacity of lubrication film. The main conclusions of this paper are as follows:

(1) The average pressure decreases with the increase of minimum film thickness. Under the condition of the same film thickness, when the medium is diesel, the average pressure is greater than the average pressure when the medium is gasoline. When the flowmeter operates in a stable condition, the maximum film thickness corresponding to the diesel medium is between 45 and 47.5 , and the maximum film thickness corresponding to the gasoline medium is between 50 and 52.5 . The clearance of flowmeter used to measure the gasoline is smaller than that of the flowmeter used to measure the diesel, so that the medium factor should be taken into account when designing the clearance value of the fuel dispenser flowmeter.

(2) It was also found that the lubrication film pressure increases with the increase of flow rate, the lubrication film load-carrying capacity is insufficient when the flow rate is less than 20L/min, resulting in increased leakage in the clearance. The flow regime of the lubrication film changes with the change of flow rate. When the flow rate is greater than 60L/min, the flow regime of clearance is transitive or turbulent flow, resulting in severe wear to the piston ring seal.

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(3) In addition, the results show that the lubrication film pressure increases with the increase of viscosity. For the same leakage clearance, the higher the viscosity, the larger load-carrying capacity of the clearance lubrication film, leading to the reduction of clearance leakage and measurement error.

However, the theoretical analysis model was developed just with the wedge effect, the viscosity and density parameters selected in the course of analysis are fixed, and the elastic deformation of the piston ring are not taken into consideration. Hence, the present conclusion is limited to that way of comparison only. It is necessary to establish an analysis model which based on elastohydrodynamic lubrication and considering the effects of viscosity and density changes in following research.

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